

Simplifying DeepSeek Deployment with Azure Container Apps



PhucNT/NashTech

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AI is changing the world

- AI is accelerating solution development
- AI is our personal customized copilot
- AI is impacting on human decision-making
- Most importantly, AI is just a **useful tool** 🖥️.



AI LLM Deployment is expensive 💰

- GPU AI or Gaming
- CPU AI or i9 equivalent
- RAM – 32GB DDR4
- Storage – 1 TB NVMe SSD
- Cooling system
- Power bills

💰 ~ 100,000,000 VND



What if we use Azure Container Apps – Consumption GPU?

- Running one Azure Container Apps serverless GPU replica that has 8 vCPU, 56 GiB memory and an NVIDIA T4 for 8 hours a day during 30 days comes to ≈ US \$383/month (≈ ₫9.8 million VND) at pay-as-you-go rates:

Meter	Billable seconds	Rate	Cost
GPU-seconds (1 × T4)	864 000	\$0.00009	\$ 77.76
vCPU-seconds (8 cores)	6 912 000 – 180 000 free grant = 6 732 000	\$0.000024	\$ 161.57
GiB-seconds (56 GiB)	48 384 000 – 360 000 free grant = 48 024 000	\$0.000003	\$ 144.07
Total (30 days)			\$ 383.40
(exchange 25 500 ₫/USD ⇒ ≈ ₫9 776 700)			

What is Ollama?

- Ollama is an open-source tool that allows you to run large language models (LLMs) directly on your local machine



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deepseek-r1

ollama run deepseek-r1

43.3M Downloads Updated 3 months ago

DeepSeek's first-generation of reasoning models with comparable performance to OpenAI-o1, including six dense models distilled from DeepSeek-R1 based on Llama and Qwen.

1.5b 7b 8b 14b 32b 70b 671b

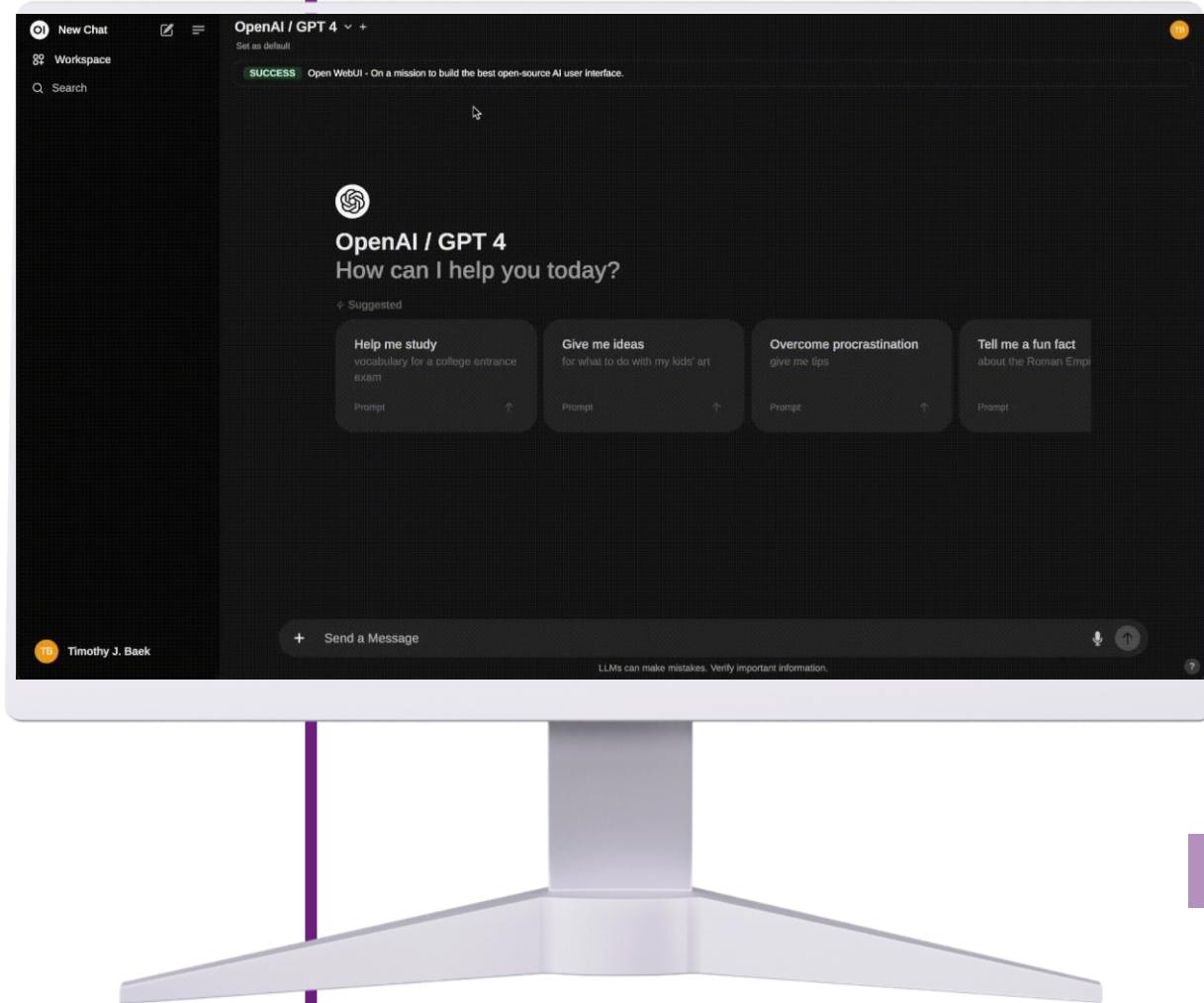
7b 29 Tags

Updated 3 months ago	0a8c26691023 · 4.7GB
model	arch qwen2 · parameters 7.62B · quantization Q4_K_M 4.7GB
params	{ "stop": ["< begin_of_sentence >", "< end_of_sentence >"], 146B
template	{{- if .System }}{{ .System }}{{ end }} {{- range \$i, \$_ := .Message- 387B
license	MIT License Copyright (c) 2023 DeepSeek Permission is hereby grante... 1.1kB

Readme



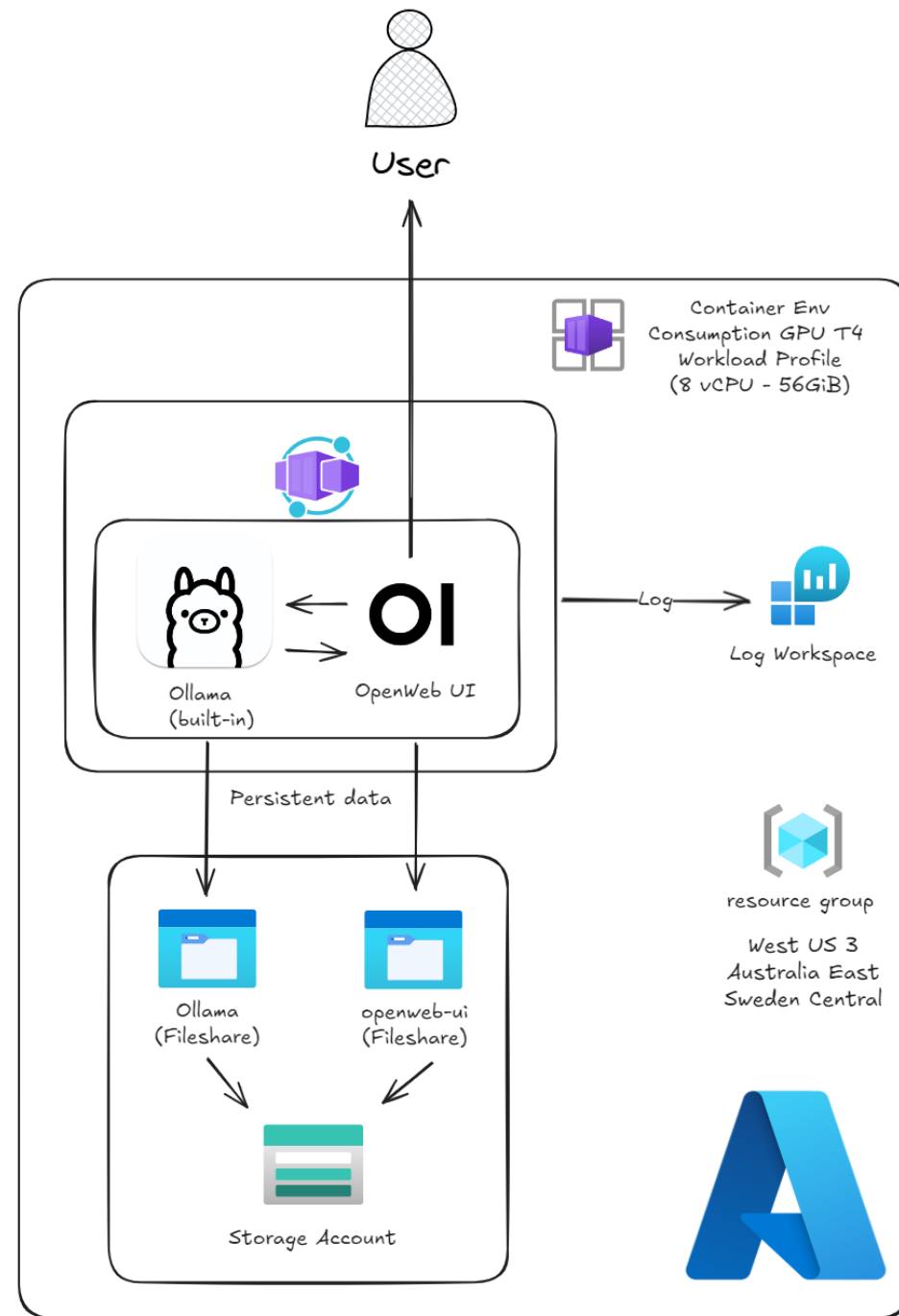
DeepSeek's first-generation reasoning models, achieving performance comparable to OpenAI-o1 across math, code, and reasoning tasks.



Open WebUI

Is an **extensible**, **feature-rich**, and **user-friendly self-hosted AI platform** designed to **operate entirely offline**.

It supports various LLM runners like **Ollama** and **OpenAI-compatible APIs**, with built-in inference engine for RAG, making it a powerful AI deployment solution.





Demo

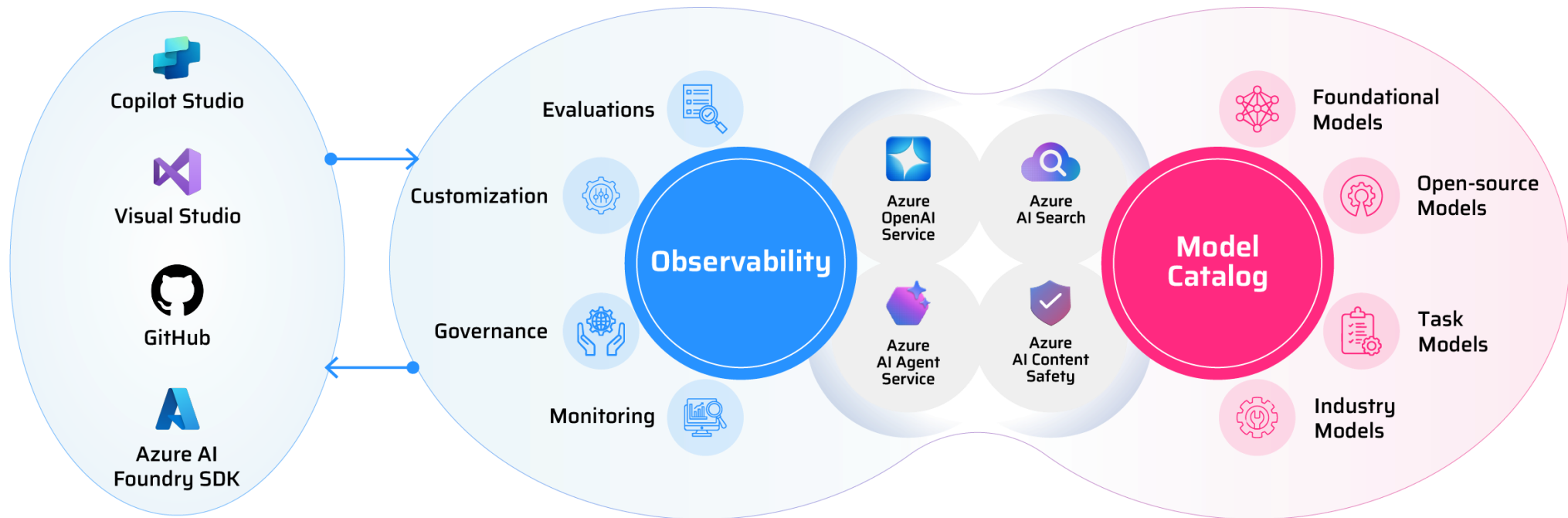
[phucnt1992/deepseek-on-aca-101](https://github.com/phucnt1992/deepseek-on-aca-101): A simple workshop to deploy deepseek to Azure Container Apps

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Alternative Solution



Azure AI Foundry



[Link](#)

Thank you

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